NUMPY PROGRAMS

1. write a NumPy program to read the marks of 5 students and find avg, highest and lowest marks
   input: (78 65 92 55 84)

main.py Visualize Run

```python
import numpy as np

marks = np.array([78, 65, 92, 55, 84])

print("Average:", np.mean(marks))
print("Highest:", np.max(marks))
print("Lowest:", np.min(marks))
```

Output

```
Average: 74.8
Highest: 92
Lowest: 55

=== Code Execution Successful ===
```

2. a sensor records temp value; negative value represent invalid readings. write a NumPy program to replace all the negative value with 0
   input: 25 -3 28 -1 30 27 -5

main.py Visualize Run

```python
import numpy as np
temp = np.array([25, -3, 28, -1, 30, 27, -5])
temp[temp < 0] = 0
print(temp)
```

Output

```
[25  0 28  0 30 27  0]

=== Code Execution Successful ===
```

3. the marks obtained by the 5 students in tests are given as NumPy array. write a program to display the students whose 2nd test mark is higher than 1st test marks
   input:
   test1: 65 80 72 90 55
   test2: 70 75 78 85 60
   output:
   students improved: [1 3 5]

main.py Visualize Run

```python
import numpy as np
test1 = np.array([65, 80, 72, 90, 55])
test2 = np.array([70, 75, 78, 85, 60])
students = np.where(test2 > test1)[0] + 1
print("Students improved:", students)
```

Output

```
Students improved: [1 3 5]

=== Code Execution Successful ===
```

4. write a NumPy program to read 3x3 matrix and display the sum of each column and each row
   input:
   1 2 3
   4 5 6
   7 8 9
   output:
   rows sum = [6 15 24]
   column sum = [12 15 18]

main.py

```python
import numpy as np
matrix = np.array([
    [1, 2, 3],
    [4, 5, 6],
    [7, 8, 9]
])
print("Rows sum =", np.sum(matrix, axis=1))
print("Column sum =", np.sum(matrix, axis=0))
```

Output

```
Rows sum = [ 6 15 24]
Column sum = [12 15 18]

=== Code Execution Successful ===
```

5. rainfall recorded over 12 months is stored in NumPy array, write a prog to find the total rainfall, avg rainfall, months having rainfall above avg
   input:
   120 85 150 200 175 90 110 160 210 180 95 130
   output:
   total rainfall: 1705.0
   avg rainfall: 142.08
   months having rainfall above avg: [3 4 5 8 9 10]

Python Online Compiler

main.py

```python
import numpy as np
rainfall = np.array([120, 85, 150, 200, 175, 90, 110, 160,
210, 180, 95, 130])
total = np.sum(rainfall)
avg = np.mean(rainfall)
months = np.where(rainfall > avg)[0] + 1
print("Total rainfall:", total)
print("Avg rainfall:", round(avg, 2))
print("Months having rainfall above avg:", months)
```

Output

```
Total rainfall: 1705
Avg rainfall: 142.08
Months having rainfall above avg: [ 3  4  5  8  9 10]

=== Code Execution Successful ===
```